MEDTOUREASY

Patient Conversion & Provider Performance Intelligence

Verified analytics report | Patient journey, provider benchmarking, financial analysis and ML evaluation

Project objective

Use patient-journey data to understand conversion and recorded drop-off, identify operational bottlenecks, compare provider outcomes, and test whether inquiry-time information can predict drop-off.

DATASET-BASED PORTFOLIO REPORT • 2026

Prepared by analysing the original Project.zip source files. Model metrics were rerun using the documented split and model settings. Descriptive findings are not causal claims.

Executive Summary

This report analyses 1,500 patient records, 30 providers, 6,025 communication-log records, and 554 treatment-outcome records. The objective is to help patient operations teams see where journeys stall, compare provider outcomes, and understand whether simple inquiry-time features can reliably predict recorded drop-off.

The recorded drop-off target is positive for 350 patients (23.3%). 554 patients (36.9%) are marked treatment-completed. Another 596 patients (39.7%) are neither flagged as drop-off nor marked treatment-completed; they should be treated as unresolved or status-uncertain, not automatically as successful conversions.

Operationally, the highest observed stage drop-off rates are Consultation (39.7%) and First Response (39.0%). The most frequently recorded drop-off reasons are No follow-up (66), Cost concern (62), and Response delay (60). These suggest actionable workflow reviews, while remaining descriptive records rather than proof of cause.

The patient-level financial fields reconcile to recorded revenue of ■248,325,245, service cost of ■136,810,889, and gross margin of ■111,514,356. These are calculated from source fields; they should still be interpreted as dataset financial measures, not audited accounts. The three machine-learning baselines were rerun: ROC-AUC ranges from 0.489 to 0.516, close to random ranking. No model is ready for deployment.

Executive conclusion: the strongest current deliverable is an operational analytics workflow and provider-review dashboard. Improve status definitions and case follow-up tracking first; revisit predictive modelling only after the target labels and features are validated.

Key numbers

Measure	Verified result	Meaning
Patient records	1,500	Patient journey dataset
Recorded drop-off	350 (23.3%)	Target label prevalence
Treatment completed	554 (36.9%)	Completed treatment flag
Status-uncertain records	596 (39.7%)	Neither completed nor flagged drop-off
Providers	30	Provider master records
Communication logs	6,025	Interaction records
Best ROC-AUC	0.516	Near-random ranking

1. Business Problem and Analytical Scope

Patients move through multiple hand-offs—from inquiry and first response to consultation, quote, treatment and follow-up. If operational teams cannot see where cases stall, they may miss follow-up opportunities, misunderstand provider performance, or confuse active cases with confirmed drop-offs.

Project questions

- At which journey stages are patient records concentrated and where is recorded drop-off most common?
- Which reasons are recorded for drop-off, and how do rates vary descriptively across lead sources and patient segments?
- How do providers compare on consultation, quote, treatment completion, follow-up, satisfaction and recorded revenue?
- What do the revenue, service-cost and gross-margin fields show?
- Can inquiry-time features predict recorded drop-off better than chance?

Scope boundary

This is a portfolio analytics project, not a clinical decision system. It does not establish medical eligibility, treatment quality, causal drivers, or audited profitability. Findings are intended to support operational investigation and measurement.

2. Data Sources and Validation

Source	Rows	Use in this report
patient_journey.csv	1,500	Patient demographics, inquiry fields, stages, flags, reason codes and financial fields
provider_master.csv	30	Provider profile, specialty, response-time reference and historical measures
communication_logs.csv	6,025	Interaction events, response-sent indicator, delay and follow-up status
treatment_outcomes.csv	554	Treatment-level outcomes, revenue, service cost, gross margin, stay and readmission fields
country_reference.csv	12 (source file)	Country reference values
data_dictionary.csv	67	Field descriptions and examples

2.1 Core data checks

- The patient journey has 1,500 unique patient IDs across 1,500 rows.
- The provider master contains 30 records; all patient provider IDs should be checked against it during refreshes.
- The treatment-outcome file contains 554 rows, consistent with the 554 patients marked treatment-completed in the journey file.
- Patient-level revenue, service cost and gross margin reconcile to treatment-outcome totals in the supplied source files.
- The report uses recorded values and does not silently substitute missing values with assumptions.

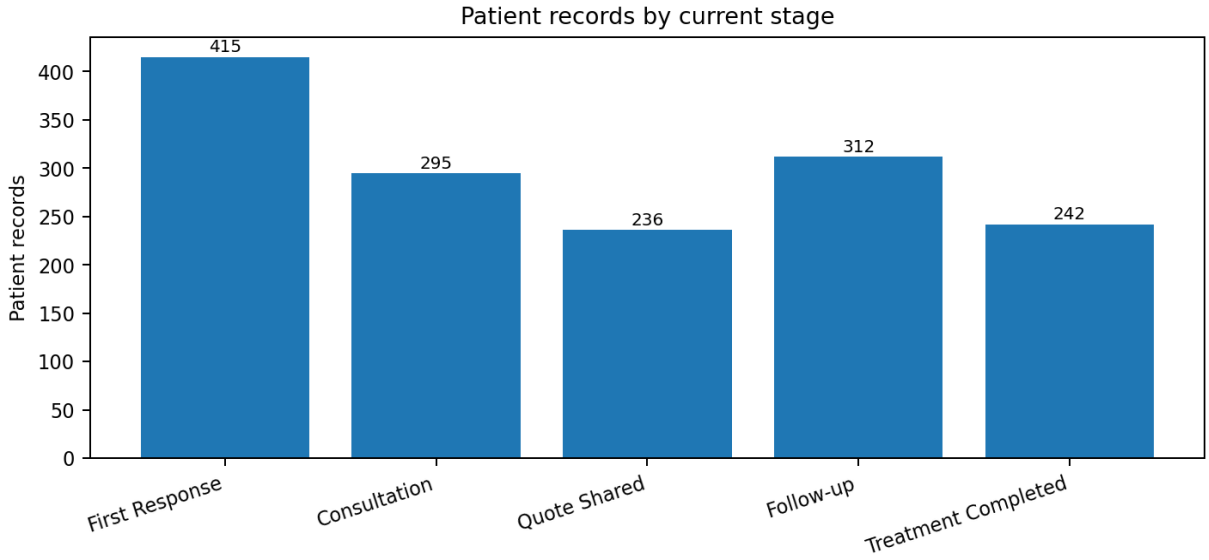
2.2 Status definition warning

drop_off_flag	Treatment not completed	Treatment completed	Total
0	596	554	1150
1	350	0	350
Total	946	554	1500

The 596 patients with drop_off_flag = 0 and treatment_completed = 0 are status-uncertain for business reporting. The target should be refined into mutually exclusive completed, active/in-progress and confirmed-drop-off states before operational risk scoring.

3. Patient Journey Analytics

3.1 Stage volumes



The largest current-stage groups are First Response (415), Consultation (295), Follow-up (312), Quote Shared (236), and Treatment Completed (242). Current stage is a snapshot field, not a complete event funnel; stage counts should not be added together as if they were sequential conversion events.

3.2 Recorded drop-off by stage

Current stage	Patients	Recorded drop-offs	Rate within stage
Consultation	295	117	39.7%
First Response	415	162	39.0%
Quote Shared	236	71	30.1%
Follow-up	312	0	0.0%
Treatment Completed	242	0	0.0%

Consultation and First Response have the highest observed stage-level recorded drop-off rates, at about 39–40%. Follow-up and Treatment Completed have zero recorded drop-offs in this cross-tab. That pattern may partly reflect stage and label construction, so zero should not be interpreted as proof that follow-up prevents drop-off.

3.3 Unresolved queue

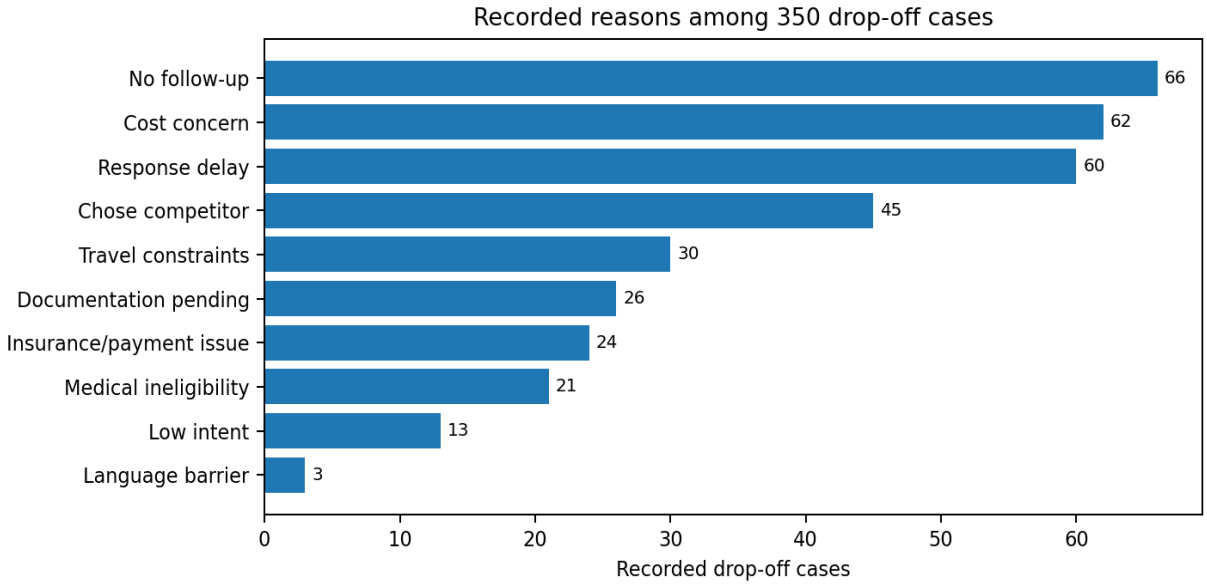
Among the 596 records that are not flagged as drop-off but are not marked treatment-completed, the saved analysis identifies 253 in First Response, 178 in Consultation and 165 in Quote Shared. These are a practical starting queue for case review once the current status and last-contact date are refreshed.

Operational implication

- Build a case queue with current stage, days since last contact, assigned owner, next action and reason for delay.
- Escalate cases without an owner or next-contact date.
- Separate active cases from confirmed drop-offs in dashboards and reporting.

4. Recorded Drop-off Reasons and Segments

4.1 Reasons recorded for the 350 flagged drop-offs



Reason	Count	Share
No follow-up	66	18.9%
Cost concern	62	17.7%
Response delay	60	17.1%
Chose competitor	45	12.9%
Travel constraints	30	8.6%
Documentation pending	26	7.4%
Insurance/payment issue	24	6.9%
Medical ineligibility	21	6.0%
Low intent	13	3.7%
Language barrier	3	0.9%

No follow-up, cost concern, and response delay are the three largest recorded categories. These categories suggest practical questions for operations teams: Was ownership assigned? Was the expected cost communicated early? Was the response-time commitment met? The reason codes are recorded classifications and should not be described as verified causal effects.

4.2 Descriptive segment differences

Selected group rates from the original data are shown below. Group sizes are included because a rate without its denominator can be misleading.

Dimension	Group	Patients	Drop-off rate
Lead source	Web Form	246	27.2%
Lead source	Email	238	18.9%
Treatment	Fertility	166	27.7%
Urgency	Urgent	305	28.9%
Budget	Low	247	27.1%
Age group	50–59	300	29.0%

These are descriptive associations, not evidence that a segment characteristic causes drop-off. They should inform service-process questions and data quality checks—not exclusionary patient decisions.

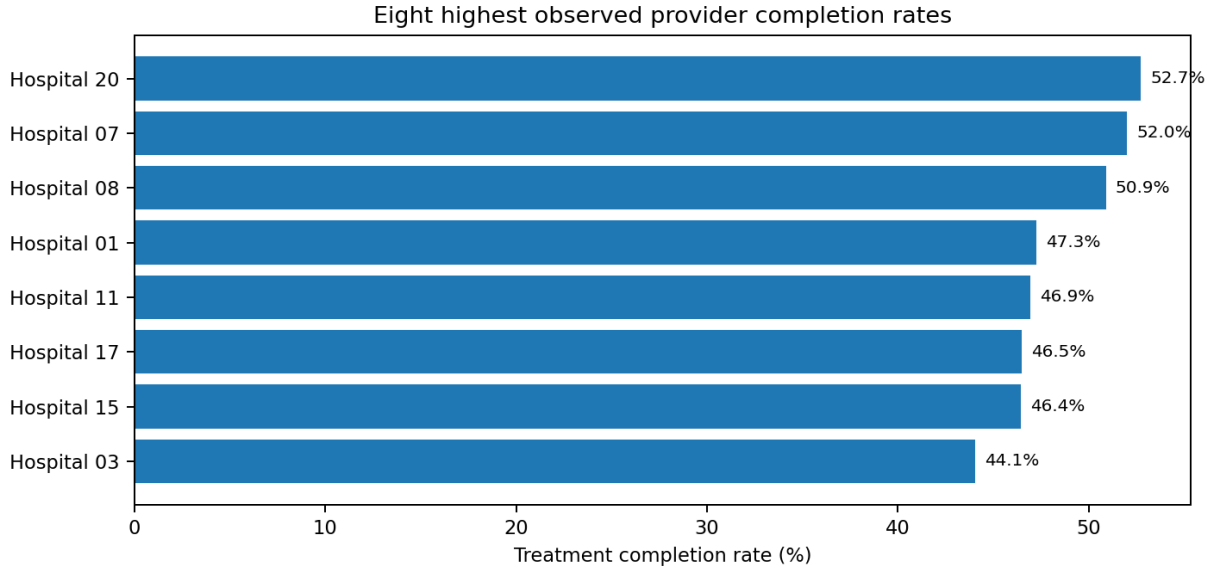
5. Provider Performance Intelligence

5.1 How providers are compared

Observed patient-level metrics are calculated from the patient journey and joined to provider master names. These observed rates are distinct from the historical conversion measures stored in provider_master.csv.

Metric	Interpretation
Patient volume	Unique patient records associated with provider
Consultation / quote rate	Share of provider-linked patient records with the respective event flag
Treatment completion	Share with treatment_completed = 1
Follow-up completion	Share with follow_up_completed = 1
Recorded drop-off	Share with drop_off_flag = 1
Revenue / service cost	Sum of recorded patient-level INR fields; gross margin = revenue minus service cost
Satisfaction	Mean available satisfaction score; interpret alongside number of scored patients

5.2 Highest observed treatment-completion rates



Provider	Volume	Completion	Drop-off	Follow-up	Revenue
Hospital 20	55	52.7%	14.5%	38.2%	■13,379,748
Hospital 07	50	52.0%	18.0%	24.0%	■12,239,804
Hospital 08	55	50.9%	12.7%	14.5%	■13,774,849
Hospital 01	55	47.3%	18.2%	36.4%	■12,202,774
Hospital 11	49	46.9%	20.4%	22.4%	■8,513,562

Hospital 20, Hospital 07 and Hospital 08 are among the strongest observed performers on treatment completion in this dataset. Hospital 04 has the largest patient volume (66), but volume alone is not a performance measure. Compare multiple metrics and review case mix before drawing conclusions.

5.3 Providers for operational review

The following providers are candidates for review because their recorded drop-off rates are relatively high in the observed data. This is a screening list—not a quality rating—and should be reviewed alongside specialty, patient mix, volume, and data completeness.

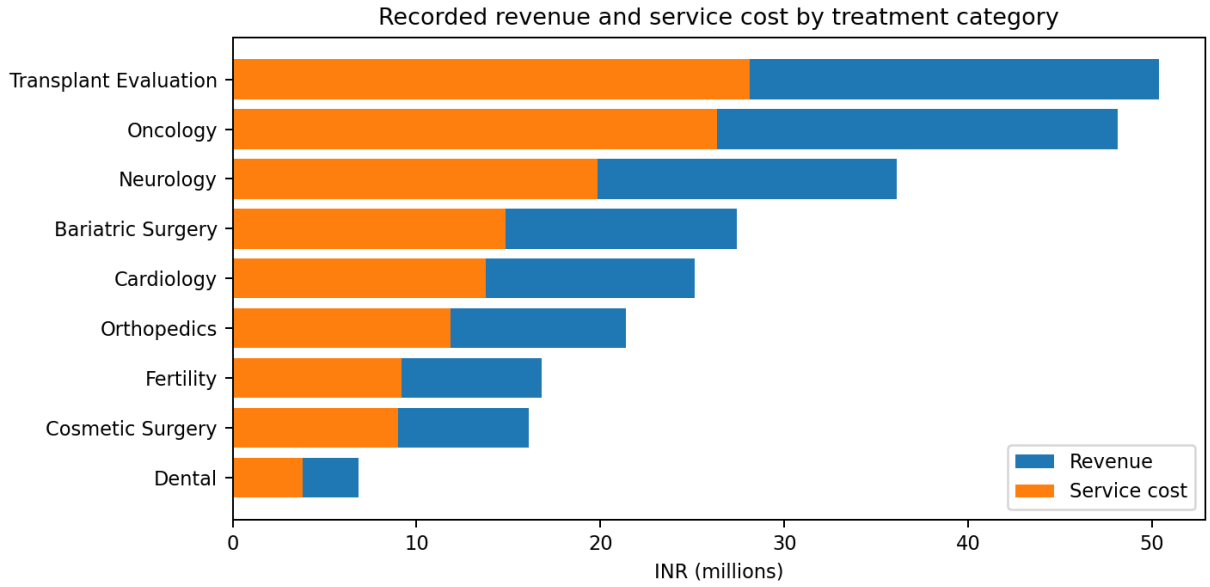
Provider	Volume	Completion	Drop-off	Follow-up	Satisfaction
Hospital 18	48	31.2%	37.5%	20.8%	7.43
Hospital 13	50	24.0%	36.0%	12.0%	7.03
Hospital 23	46	23.9%	34.8%	8.7%	7.27
Hospital 10	41	36.6%	31.7%	14.6%	7.54
Hospital 05	48	27.1%	31.2%	10.4%	7.12
Hospital 26	46	39.1%	30.4%	17.4%	7.58
Hospital 30	50	36.0%	30.0%	22.0%	7.51
Hospital 29	38	26.3%	28.9%	18.4%	6.85

Suggested review: inspect response-time distributions, follow-up ownership, quote progression and recorded reasons for stalled cases. Do not assume the provider caused the outcome; the dataset is observational.

6. Revenue and Financial Analysis

The original patient-journey fields contain recorded revenue and service cost. Totals reconcile with the treatment_outcomes.csv file: revenue ■248,325,245, service cost ■136,810,889, and gross margin ■111,514,356. The calculated gross margin is revenue minus service cost; it is not operating profit or net profit.

6.1 Revenue and cost by treatment category



Treatment category	Patients	Revenue	Service cost	Gross margin
Transplant Evaluation	161	■50,367,261	■28,124,580	■22,242,681
Oncology	187	■48,123,276	■26,348,999	■21,774,277
Neurology	174	■36,111,916	■19,854,602	■16,257,314
Bariatric Surgery	162	■27,423,142	■14,855,857	■12,567,285
Cardiology	158	■25,108,555	■13,771,132	■11,337,423
Orthopedics	156	■21,397,906	■11,848,685	■9,549,221
Fertility	166	■16,800,212	■9,171,244	■7,628,968
Cosmetic Surgery	180	■16,126,088	■9,018,062	■7,108,026
Dental	156	■6,866,889	■3,817,728	■3,049,161

Financial interpretation is limited to the fields supplied. The report does not infer other operating expenses, taxes, overhead allocation, or net profit. Treat these as dataset financial measures, not audited financial statements.

7. Machine-Learning Evaluation

7.1 Prediction task and feature timing

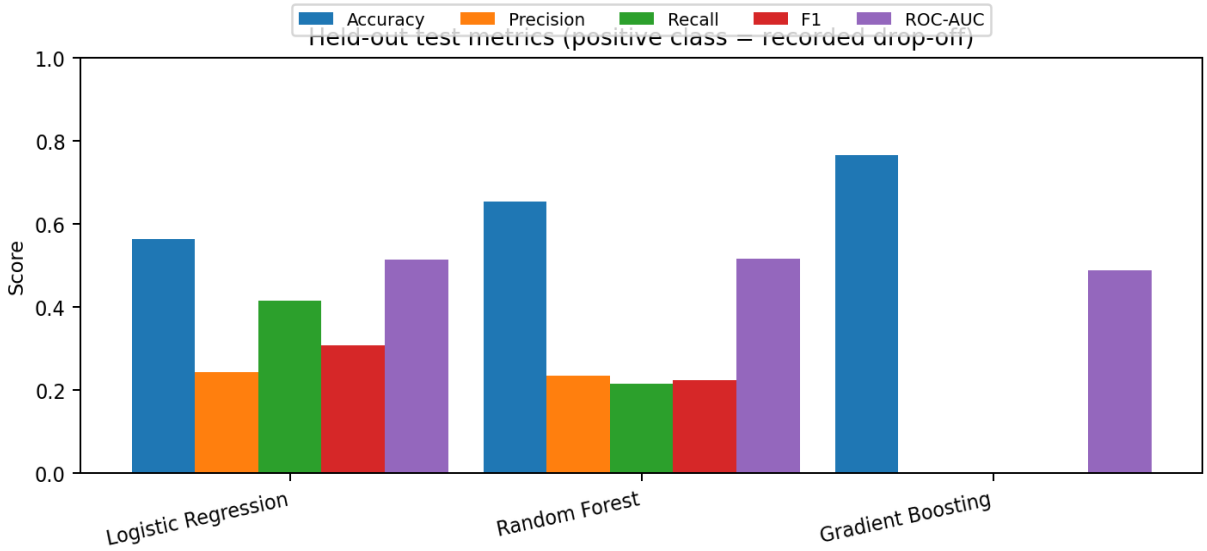
The task is to predict the recorded drop_off_flag from features intended to be available at initial inquiry: country, age_group, gender, treatment_category, lead_source, budget_band and medical_urgency. Later journey fields—such as response time, consultation booked, quote shared, follow-up completion, treatment completion and current stage—were excluded to reduce temporal leakage.

The label itself remains imperfect: target 0 includes both treatment-completed patients and 596 patients without completed treatment. A better operational target should separate completed, active/in-progress and confirmed drop-off.

7.2 Reproducible evaluation design

- 1,500 patient records; positive label 350 and negative label 1150.
- Stratified 80/20 split using random_state=42: {len(Xtr):,} training rows and {len(Xte):,} held-out test rows.
- Preprocessing: most-frequent imputation and one-hot encoding in a pipeline.
- Models: Logistic Regression (balanced class weights), Random Forest (300 trees, min_samples_leaf=10, balanced class weights), Gradient Boosting (100 estimators, learning_rate=0.05, max_depth=2).
- Metrics: accuracy, precision, recall, F1-score and ROC-AUC.

7.3 Test-set model comparison



Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.563	0.244	0.414	0.307	0.514
Random Forest	0.653	0.234	0.214	0.224	0.516
Gradient Boosting	0.767	0.000	0.000	0.000	0.489

All ROC-AUC values are close to 0.5, indicating near-random ranking. Gradient Boosting predicts all 300 test records as non-drop-off: accuracy is 76.7% because 230 of 300 are negative, but recall and F1 for the positive class are zero. Accuracy alone is misleading here.

7.4 Confusion matrices and model decision

Model	TN	FP	FN	TP	Interpretation
Logistic Regression	140	90	41	29	Finds some drop-offs but raises many false alerts.
Random Forest	181	49	55	15	Misses most actual drop-offs.
Gradient Boosting	230	0	70	0	Predicts no drop-offs in test set.

For Logistic Regression, recall is 29/70 = 41.4%, but precision is 29/(29+90) = 24.4%. It misses 41 of the 70 actual test-set drop-offs and raises 90 false-positive alerts. Random Forest identifies 15 of 70 actual drop-offs. None is suitable for automated patient-level prioritisation.

7.5 Model conclusion

No model is recommended for production. The results suggest that the selected inquiry-time features and current label do not provide enough predictive signal for a reliable classifier. A future experiment should first improve labels and add validated operational context, then compare against simple baselines and evaluate precision-recall trade-offs, calibration and alert workload.

7.6 Why the previous handwritten values differed

The models were rerun directly from the supplied patient_journey.csv using the feature list, 80/20 stratified split and stated model settings. Gradient Boosting accuracy is 0.767 (76.7%), not 96.7%; the confusion matrix is [[230, 0], [70, 0]]. The response-time coefficient noted in the handwritten pages was not used as a correlation claim in this report. A Pearson correlation must be between –1 and +1; any value outside that range is a different statistic or a calculation error.

8. Recommendations and Implementation Roadmap

Priority	Action	Measure of progress
1 — Immediate	Define mutually exclusive patient states: completed, active/in-progress, confirmed drop-off.	Status coverage; count of ambiguous records
2 — Immediate	Create unresolved-case queue with stage, last-contact date, owner and next action.	Open cases by stage; overdue cases; owner coverage
3 — Immediate	Introduce response-time targets and escalation for overdue first responses.	Median / 90th percentile response time; SLA attainment
4 — High	Require structured reason codes for stalled and lost cases.	Reason-code completeness; reason distribution by stage/provider
5 — High	Review providers with high drop-off and low completion using case-mix context.	Provider review log; trend in completion and follow-up
6 — High	Keep revenue, service cost and gross margin definitions documented.	Reconciled totals; finance sign-off on definitions
7 — Next iteration	Rebuild ML only after target labels are improved.	Recall, precision, PR-AUC, calibration and alert workload
8 — Next iteration	Track interventions prospectively and evaluate change against a defined baseline.	Before/after trend and documented evaluation design

30–60–90 day roadmap

Period	Focus	Deliverable
Days 1–30	Definitions, data checks, unresolved queue, follow-up ownership	Data dictionary and operational dashboard v1
Days 31–60	Provider review, response-time SLA, structured reasons	Monthly provider action log
Days 61–90	Measure interventions, refine labels, retest baseline models	Evaluation report and ML go/no-go decision

9. Dashboard Specification

The dashboard should let operations teams move from overall KPIs to the specific stage, provider, or unresolved case group requiring action. It should show source definitions and sample sizes, not just attractive visuals.

Overview

- KPI cards: patient records, recorded drop-off rate, treatment completion, status-uncertain count, follow-up completion and recorded revenue.
- Stage distribution and drop-off rate chart; label as current-stage snapshot.
- Filters: country, treatment category, lead source, urgency and provider.

Provider performance

- Provider table with patient volume, consultation, quote, completion, follow-up, drop-off, revenue, service cost, gross margin and satisfaction.
- Drill-down into stage distribution and recorded drop-off reasons.
- Show volume and number of satisfaction scores beside rates.

Patient journey

- Recorded reason distribution and unresolved queue by stage.
- Separate completed, active/in-progress and confirmed drop-off states once the data definition is agreed.
- Highlight overdue next actions rather than labelling individuals with unreliable ML risk scores.

Model evaluation

- Display all five metrics, confusion matrices, test-set size and target definition.
- Clearly mark current models as not production-ready.
- Track any future experiment against baseline metrics and threshold workload.

Governance

Each KPI should have a written definition and denominator. Use role-based access for patient-level information, avoid unnecessary personal data, and date-stamp each refresh. Financial charts should distinguish revenue, cost and gross margin explicitly.

10. Limitations and Next Steps

- The drop-off target is not mutually exclusive with the treatment-completion status because 596 records have neither completed treatment nor a positive drop-off flag.
- The dataset has 1,500 patients and 350 positive labels; this limits the reliability of subgroup analysis and predictive modelling.
- Model ROC-AUC is near random and the positive-class precision/recall are weak.
- Provider comparisons are observational and may reflect specialty, case mix, country, patient volume or data completeness.
- Recorded drop-off reasons may be incomplete or reflect how the CRM captured outcomes.
- Financial fields reconcile within the supplied files, but they are not independently audited accounting records.
- The report does not claim causation or a successful production ML model.

Appendix A — Metric definitions

Metric	Definition in this report
Recorded drop-off rate	Records with drop_off_flag = 1 divided by patient records in the stated group.
Treatment completion rate	Records with treatment_completed = 1 divided by patient records in the stated group.
Follow-up completion rate	Records with follow_up_completed = 1 divided by patient records in the stated group.
Gross margin	Recorded revenue minus recorded service cost. Does not include other operating expenses.
Precision	TP / (TP + FP); share of predicted drop-offs that are actual drop-offs.
Recall	TP / (TP + FN); share of actual drop-offs identified.
F1-score	Harmonic mean of precision and recall.
ROC-AUC	Ranking ability across thresholds; 0.5 is approximately random.
Pearson correlation	Must lie between –1 and +1.

Appendix B — Interview-ready project summary

“I analysed a healthcare patient-journey dataset of 1,500 records across 30 providers, along with communication and treatment-outcome data. I measured stage-level drop-off, reviewed recorded reasons, compared provider outcomes and reconciled revenue, service cost and gross margin. I also tested Logistic Regression, Random Forest and Gradient Boosting using inquiry-time features. Their ROC-AUC scores were close to 0.5, so I did not present them as production-ready. The main business recommendation is to clarify patient status definitions and improve case follow-up visibility before attempting reliable predictive prioritisation.”

Source and reproducibility note

This report was generated from the original Project.zip files: patient_journey.csv, provider_master.csv, communication_logs.csv, treatment_outcomes.csv, country_reference.csv and data_dictionary.csv. Provider metrics, financial totals and model results were recalculated from those source files. The report should be refreshed whenever the source data or business definitions change.

End of report